

Need for NAWODHAN

Kerala has a total cropped area of 25,23,014.31 hectare of which more than 85% of the area is covered under perennial crops and only 15% of the area is utilised for food crops. It is estimated that 1,03,334 hectares of fallow land is available in the State, of which 53,914 Ha is classified as current fallow and 49,420 Ha as permanent fallow. The current vegetables' requirement of Kerala is 30 lakh Metric Tonne (MT) per year while the production is 17.24 lakh MT per year indicating a gap of 12.76 lakh MT which is met from the neighbouring States. Similarly, the current fruit requirement in the State is 40 lakh MT per year with an annual production of around 26 lakh MT per year.

Kerala faces fruit and veggie scarcity due to resource scarcity, labor shortage, and lack of interest over farming activities. Dependence on external wholesale markets makes price regulation difficult. Additionally, labour shortages due to high wages and migration to urban areas have exacerbated the agricultural challenges. Underutilized lands in Kerala can be leveraged to boost domestic production. Land fragmentation is another significant issue. Small and scattered landholdings make it difficult to implement large-scale farming techniques and mechanization, which are essential for improving productivity and sustainability. Fallow lands and inter-cropping areas can double farmers' income and alleviate scarcity. Effective utilization of these lands can address the shortage and stabilize prices.

The increasing demand for fruits and vegetables in both domestic and international markets further underscore the need for Kerala to enhance its agricultural productivity and self-sufficiency. High value crops present significant opportunities in Kerala due to the state's favourable climate and soil, technological advancements in farming, and growing market demand for quality produce. Government support through policies and incentives further boosts this sector. Sustainable practices and innovations like hydroponics and precision farming enhance productivity and efficiency, leading to higher income for farmers and contributing to economic growth.

In view of the above factors, Government has accorded sanction in-principle for the project titled "New Agriculture Wealth Opportunities-Driving Horticulture and Agribusiness Networking (NAWO-DHAN) for leveraging the unutilised /underutilised land resources in Kerala for commercial farming practices that include horticulture, inter-cropping, mushroom cultivation, apiculture, crop husbandry, allied sectors, hydroponics, protected cultivation, precision-farming and hi-tech farming. Through initiatives like NAWODHAN, the increase in fruit and vegetable production can effectively meet the growing demand in both domestic and international markets. This programme outlines a comprehensive plan to cultivate high-value crops for international markets, leveraging the expertise of Kerala Agro Business Company (KABCO) and the resources available under the NAWODHAN scheme. Kerala Agro

Business Company (KABCO) has been nominated by the Government of Kerala as the Special Purpose Vehicle (SPV) to implement NAWODHAN.

Vision

- To harness the potential of vacant cultivable land for hi-tech commercial farming
- To identify and cultivate high-value fruits, vegetables and commercial crops suitable for international markets
- To establish a sustainable and profitable farming model by advocating for the adoption of modern and scientific farming techniques.
- To boost the income of farmers and agripreneurs by promoting high-value agricultural produce through export-oriented agriculture.

KABCO as SPV

In the 2016-17 State Budget, the Government of Kerala announced the establishment of a Company in the Cochin International Airport Authority Ltd. (CIAL) model. Building on this initiative, the 2022-23 budget proposed the creation of the Kerala Agro Business Company (KABCO) with a capital of Rs. 100 crores. This company aims to enhance the markets for quality-assured, value-added agricultural products, and an amount of Rs. 20 crores have been earmarked for the purpose.

The formation of KABCO is intended to establish a vibrant agricultural marketing system in Kerala, thereby increasing farmers' income by focusing on value addition and processing.

KABCO has been designated as the Special Purpose Vehicle for implementing the NAWO-DHAN project (G.O.(Ms) No. 74/2024/AGRI) under the supervision of the Department of Agriculture Development & Farmers' Welfare. KABCO will liaise with various stakeholders to inventorize the cultivable land available with different departments, farms, State and Central PSUs, private individuals, and institutions. Acting as an aggregator, KABCO will develop these lands into hi-tech agricultural farms of various sizes. Farmer groups will be engaged for cultivation through Service Level Agreements, treating "Farming as a Service" without transferring any interest in the land. KABCO will also develop forward linkages, including export-market integration, ensuring business-intelligence-driven cropping that aligns production with market demand.

- KABCO will acquire underutilized land from government bodies, PSUs, and private landholders, and allocate it to farmers, FPOs, and startups through a transparent process.
- KABCO will coordinate stakeholders, manage logistics, and ensure regulatory compliance, providing administrative and technical support to streamline farming activities.

- KABCO will connect farmers with domestic and international buyers, identify market opportunities, negotiate trade agreements, and organize buyer-seller meets to maximize returns.
- KABCO will conduct training programs and workshops on modern farming techniques, quality control, and export procedures to enhance participants' skills and knowledge.
- KABCO will guide participants through certification processes like Global GAP, organic certification, and HACCP to meet international market standards.
- KABCO will earn a commission on produce sales by negotiating competitive prices and managing the supply chain efficiently.
- KABCO will charge a fee for aggregated land provided to participants, ensuring affordability while covering administrative costs.
- KABCO will charge for facilitation services like training, certification, and logistical support.
- KABCO will manage funds and grants from various sources to cover operational costs and expand initiatives.
- KABCO will form strategic partnerships for joint ventures, sponsorships, and funding opportunities to support and grow its activities.

Global Scenario

Highlights of the Global F&V Market

Fruits and vegetables have gained considerable commercial importance all over the World, contributing significantly to the economics of many countries. They are important supplement to the nutritional requirements in the human diet as they provide the essential minerals, vitamins and fibre required for maintaining health.

Revenue in the fresh fruits market is projected to reach \$726.20 billion in 2024, with an expected annual growth rate of 6.42% from 2024 to 2029. Globally, India is anticipated to generate the highest revenue at \$116 billion in 2024. On a per capita basis, revenues of \$93.72 are expected in 2024. The market volume is forecasted to reach 311.10 billion kg by 2029, with a volume growth of 4.0% in 2025. The average volume per person in the fresh fruits market is expected to be 34.3 kg in 2024

Revenue in the Fresh Vegetables market amounts to US\$813.20bn (INR 67.47 Lakh Crore) in 2024. The market is expected to grow annually by 6.80% (CAGR 2024-2029).

Highlights of the F&V Market in India

- The processed fruits and vegetables including of pulses exported to be Rs. 20,623.70 Crore/ USD 2,488.72million which comprised of processed vegetables including of pulses Rs. 14,339.94Crore/ USD 1,730.79million and processed fruits and juices Rs. 6,283.76crores/ USD 757.93million in 2023-24.

- Grapes, Pomegranates, Mangoes, Bananas, and Oranges account for the larger portion of fruits exported from the country while Onions, Mixed Vegetables, Potatoes, Tomatoes, and Green Chilly Contribute largely to the vegetable export basket.
- Major destinations for the Indian Fresh Fruits and Vegetables are Bangladesh, UAE, Netherland, Nepal, Malaysia, Iraq, U.K, Sri Lanka, Iran, Oman and Saudi Arabia.
- Major destinations for the Indian Processed Fruits and Vegetables are U.S.A, UAE, Bangladesh, U.K, Saudi Arab, China and Netherland.

Vegetable Export from India to Dubai:

- Vegetable export from India to Dubai are rising as the Middle East's desire for fresh, high-quality food grows. India is well-known for its broad selection of fruits and vegetables, mainly tropical fruits, which are in great demand in Dubai.
- In 2023, India's vegetable exports will be worth approximately \$865.24 million (INR 7,100 Crores) On the other hand, fresh fruit exports from India totalled 770.70 USD million (INR 6,300 Crores).

Though India's share in the global market is nearly 1% only, there is increasing acceptance of horticulture produce from the country. This has occurred due to concurrent developments in the areas i.e. state-of-the-art in cold chain infrastructure and quality assurance measures. Apart from large investments pumped in by the private sector, the public sector has also taken the initiative with APEDA for setting up several Centers for Perishable Cargoes and integrated post-harvest handling facilities in the country. Capacity-building initiatives at the farmers, processors and exporters levels have also contributed towards this effort.

Kerala's Scenario

Kerala is at the leading edge of India's structural transformation. The State's economy is primarily driven by services (66%), and the primary sector accounts for only 12% of GSDP. This may be barely 50% of that in other states, but with potential to increase. Kerala's agricultural sector, while accounting for a smaller share of its GDP compared to other industries, plays a crucial role in its socio-economic landscape. With over 1.3 million farmers engaged in cultivation, it contributes significantly to food security, rural livelihoods, and environmental sustainability.

Kerala's varied agro-climatic conditions make it an ideal region for cultivating high-value fruits and vegetables, presenting a significant opportunity for export expansion. With the global demand for exotic and premium produce on the rise, Kerala can capitalize on this trend to establish itself as a leading exporter. The major Cash crops cultivated are

coconut, rubber, pepper, coffee, and cashew nuts dominate the agricultural landscape, accounting for over 60% of the cropped area. And other food crops such as rice, tapioca, pulses, and vegetables constitute a proportion but are vital for local food security.

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Vegetable & fruits farming has recently been regarded as one of India's fastest growing and most profitable sectors. This is primarily due to the increased demand for fresh and frozen vegetable/fruits products worldwide.

Kerala's Advantage

- Diverse Agro-climatic conditions
- Diversified cropping pattern
- High literacy rate
- Strong educational infrastructure
- High Standard of living
- Rich heritage and culture
- Significant investments in world class ports, logistics & supply chain infrastructure.
- Largest producer of several agri commodities.
- Highly skilled manpower pool.
- Agro and Eco-Tourism hub
- Proactive Government policies.
- New technology in F&V processing
- Cold chain & packhouses- Farm level, logistics, end product storage and at point of retail.
- Packaging technology
- Availability Food testing labs with latest equipment and technology

NAWO-DHAN Scope

NAWO-DHAN aims to provide comprehensive support to farmers, FPOs, Agripreneurs, consumers and Landowners in Kerala, fostering a vibrant agribusiness ecosystem.

The scheme's components will facilitate:

Technical Assistance

NAWO-DHAN offers technical assistance to farmers, FPOs, and agripreneurs through training, crop selection and soil management guidance, sustainable farming support, and regular farm visits by experts from KAU, ICAR-KVK, etc., enhancing skills, yields, eco-friendly practices, and overall farming efficiency.

Demand driven crop models

Demand-driven crop models, executed with the help of startups, optimize production based on market demand, ensuring efficient resource use and maximizing profits.

Financial Support

NAWO-DHAN offers financial support through credit access, subsidies, and insurance, helping farmers invest in technology and infrastructure, manage finances, reduce risks, and boost productivity.

Market Access

NAWO-DHAN connects farmers with domestic and international markets, offering market research, branding support, and partnerships, which increase revenue, reduce waste, and promote economic growth.

Technology Adoption

NAWO-DHAN promotes precision farming, hydroponics, and aquaponics by providing guidance and training, enhancing productivity, efficiency, and sustainability while reducing costs and environmental impact.

Infrastructure Development

NAWO-DHAN supports infrastructure development, offering guidance and financing for greenhouses and vertical farms, enhancing productivity, efficiency, and sustainability while improving crop yields and quality.

Business Planning and Management

NAWO-DHAN offers guidance on business planning, marketing, and management, helping farmers develop strategies, manage finances, and improve productivity, efficiency, and sustainability.

I. Greenhouses

Greenhouses are climate-controlled structures that enable year-round production of high-value crops like delicate fruits, vegetables, and flowers. They provide optimal growing conditions, increasing yields and quality while promoting energy-efficient and sustainable farming practices. This initiative meets the high demand for greenhouse-grown produce in local markets, ensuring a consistent and reliable supply of fresh produce throughout the year.

II. Hydroponics Farming

Hydroponic farming uses nutrient-rich solutions for soilless crop cultivation, increasing yields while reducing water and land use. Suitable for various crops and environments, it reduces waste and increases crop density, promoting efficient and sustainable farming. This method meets the high demand for fresh, high-quality produce in local markets and restaurants, minimizing environmental impact.

III. Precision Farming

Precision farming uses advanced sensors and drones to optimize crop growth and monitoring. This data-driven approach increases yields, reduces waste, and enhances efficiency and sustainability. Suitable for large-scale farming, it allows precise control over growing conditions, reduces chemical usage, and meets the demand for eco-friendly produce in local and export markets.

IV. Aquaponics

Aquaponics is a sustainable farming method that combines aquaculture (raising fish) and hydroponics (growing plants without soil). Fish waste provides organic nutrients for the plants, while the plants help filter and clean the water for the fish. This symbiotic system conserves water and reduces the need for chemical fertilizers. It is ideal for urban and small-scale farming. Aquaponics promotes efficient resource use and produces both fish and vegetables.

V. Vertical Farming

Vertical farming uses space-efficient, vertically stacked layers to grow crops, increasing yields and reducing land use. It provides optimal conditions for higher yields and quality, suitable for urban and indoor farming. This method meets the demand for fresh, locally grown produce year-round, ensuring a consistent supply regardless of weather conditions.

VI. Aeroponics System

Aeroponics uses misting systems for soilless cultivation, increasing yields and reducing water consumption. Suitable for various crops and environments, it reduces land use

and increases crop density. This method provides precise nutrient delivery, resulting in higher quality produce and more efficient, sustainable farming.

VIII. Micro Greens

Microgreens, nutrient-dense versions of leafy greens and vegetables, provide a sustainable source of fresh produce. They offer increased yields and reduced land use, ideal for small-scale and urban farming. High in demand for their flavour and nutrition, microgreens require minimal space and resources, making them perfect for local markets and restaurants.

IX. Other High-Value Crops

Specialty crops like organic produce, herbs, and spices offer a sustainable source of fresh, unique products. Ideal for small-scale farming, they increase yields and reduce waste. These high-demand crops command premium prices in local and export markets, providing profitability and promoting sustainable, diverse farming practices.

High value crops

Fruits: Avocado, Dragon Fruit, Blueberries, Passion Fruit, Mangosteen, Pomegranate, and Kiwi.

Vegetables: Asparagus, Bell Peppers, Broccoli, Cherry Tomatoes, Zucchini, Artichokes, and Sweet Corn.

Proposed High-Value cultivation practices

Organic Leafy Greens

- **Income:** ₹3,00,000 per acre per year
- **Expenditure:** ₹1,50,000 per acre per year
- **Benefits:** Healthier produce without synthetic pesticides, high demand in markets, requires minimal investment in infrastructure.

Hydroponic Tomatoes

- **Income:** ₹4,00,000 per acre per year
- **Expenditure:** ₹2,00,000 per acre per year
- **Benefits:** Water-efficient, high yield and quality, but requires significant infrastructure investment.

Vertical Farming Cucumbers

- **Income:** ₹6,00,000 per acre per year
- **Expenditure:** ₹3,00,000 per acre per year

- **Benefits:** Space-efficient for urban farming, consistent quality, but needs significant infrastructure investment.

? **Aquaponic Farming**

- **Income:** ₹5,00,000 per acre per year
- **Expenditure:** ₹2,50,000 per acre per year
- **Benefits:** Integrated fish and plant cultivation, high productivity, minimal infrastructure investment.

? **Aeroponics Herbs**

- **Income:** ₹7,00,000 per acre per year
- **Expenditure:** ₹3,50,000 per acre per year
- **Benefits:** Water-efficient and fast-growing, requires minimal infrastructure.

? **Micro Greens**

- **Income:** ₹8,00,000 per acre per year
- **Expenditure:** ₹4,00,000 per acre per year
- **Benefits:** Quick production, suitable for small spaces, high market demand.

? **Specialty Spices**

- **Income:** ₹9,00,000 per acre per year
- **Expenditure:** ₹4,50,000 per acre per year
- **Benefits:** High market and export demand, premium pricing, minimal infrastructure investment needed.

Source: From various hi-tech Cultivators

Main Programme of the project

1. **Selection of Available Cultivable Land**

Identify and secure suitable farmland through a transparent tender process to ensure fair competition and optimal use of land.

2. **Selection of Farmer/Farmer Group**

Choose suitable farmers or farmer groups who are committed to participating in the program.

3. **Handholding for Availing Credit Facilities**

Help farmers access loans and financial support by guiding them through the application process for securing funding.

4. **Matchmaking**

Pair farmers with suitable land and resources to optimize their farming potential.

5. **Providing Crop Insurance Support**

Offer crop insurance to protect farmers from crop failures and financial losses, ensuring security and risk management.

6. **Providing Technical Guidance**

Provide farmers with expert advice and technical support to improve their skills, knowledge, and crop productivity.

7. **Existing Scheme Convergence**

Align with existing government agricultural schemes and utilize additional resources to enhance the overall project impact.

These steps aim to support farmers through technical, financial, and operational means to improve agricultural success.